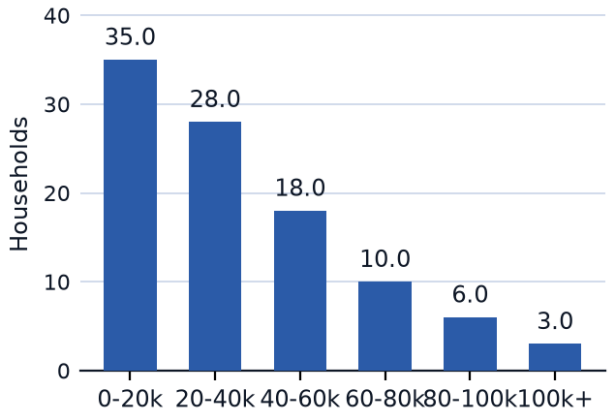
Describing Distributions & Linear Regression

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.03

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

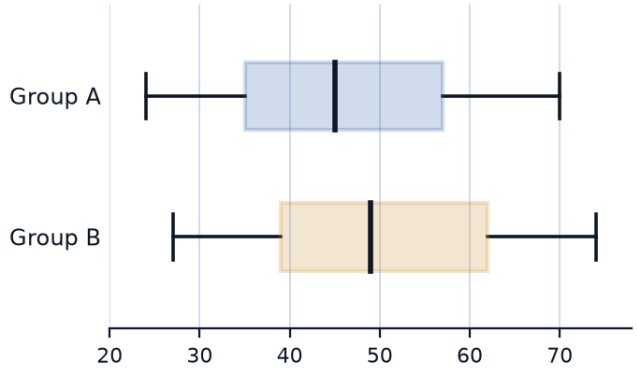
1 [Foundational] • Target: Describing shape, center, and spread

A histogram of household incomes has a long tail stretching toward high values, with most households clustered at lower incomes. Describe the shape of this distribution and state which measure of center (mean or median) is more appropriate to report, with justification.



2 [Foundational] • Target: Describing shape, center, and spread

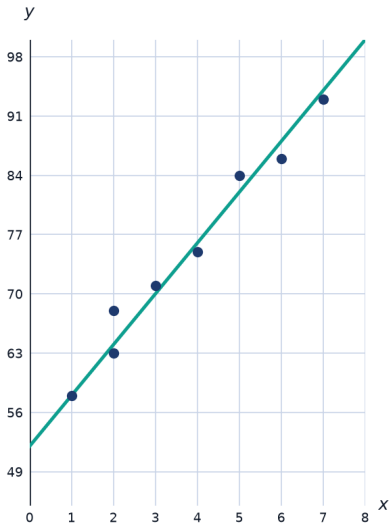
Describe what it means for a distribution to be symmetric, and state the relationship between the mean and median in that case.



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3 [Foundational] • Target: Scatterplots and correlation

The scatterplot shows hours studied (x) versus test score (y) for 8 students, along with the least-squares regression line $\hat{y} = 52 + 6x$.



a Based on the scatterplot, describe the direction and strength of the association between hours studied and test score.

b Use the regression equation to predict the test score for a student who studies 4.5 hours.

4 [Foundational] • Target: Interpreting the correlation coefficient

Explain what a correlation coefficient of $r = 0.85$ indicates about a linear relationship, and explain why r alone cannot confirm that x causes y.

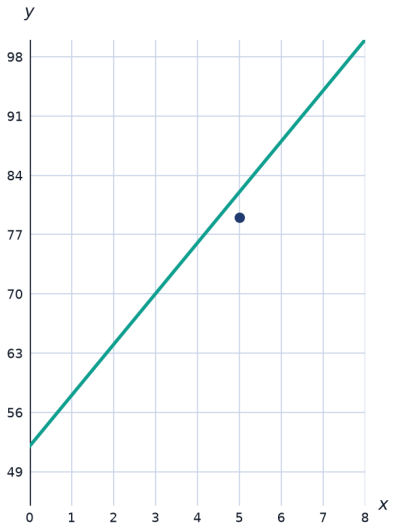
5 [Foundational] • Target: Interpreting the correlation coefficient

A data set has correlation coefficient $r = -0.4$. Describe the direction and strength of the linear relationship.

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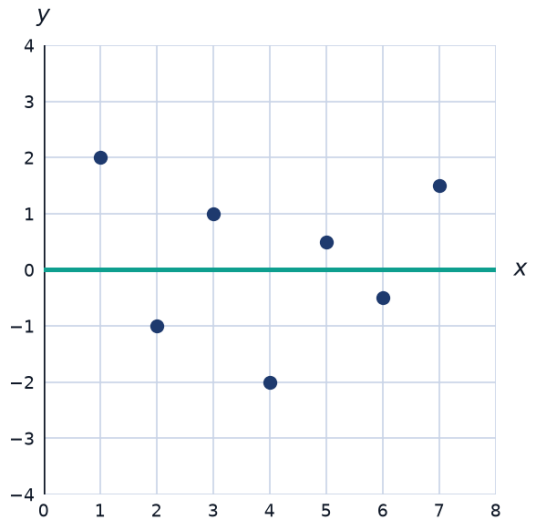
6 [Foundational] • Target: Least-squares regression and residuals

Using the regression equation $\hat{y} = 52 + 6x$ from earlier, find the residual for a student who studied 5 hours and actually scored 79.



7 [Foundational] • Target: Least-squares regression and residuals

Explain what it means for a residual plot to show "no clear pattern," and why this supports using a linear model for the data.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Coefficient of determination

For the study-hours regression, $r = 0.85$. Find r^2 and interpret its meaning in context.

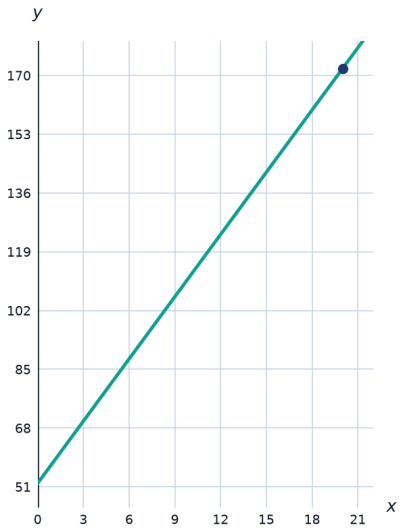
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9 **[Intermediate]** • **Target: Influential points and outliers in regression**

A data point has an unusually large x-value far from the rest of the data, and removing it noticeably changes the slope of the regression line. Explain why this point is called influential, and contrast it with a point that has a large residual but does not change the line much when removed.

10 **[Intermediate]** • **Target: Extrapolation**

Using the regression $\hat{y} = 52 + 6x$ (fit using data from students who studied between 1 and 7 hours), explain why predicting a test score for a student who studied 20 hours would be unreliable.



11 **[Intermediate]** • **Target: Least-squares regression and residuals**

Using the regression model $\hat{y} = 38.0 + 0.55x$, a student who completed 80% of homework scored 86 on the exam. (a) Calculate the residual. (b) Explain what a random scatter of points around 0 on a residual plot indicates about model adequacy.

12 **[Intermediate]** • **Target: Coefficient of determination**

A cereal manufacturer places a collectible card in each box: 20% of boxes have Card A, 30% have Card B, and 50% have Card C. Design a simulation using random digits to estimate the probability that a customer must buy at least 6 boxes to collect all three cards.

13 **[Intermediate]** • **Target: Influential points and outliers in regression**

Level 1 · FoundationsTwo independent events E and F satisfy $P(E) = 0.35$ and $P(F) = 0.60$. Find: (a) $P(E \cap F)$, (b) $P(E \cup F)$, and (c) $P(E \mid F)$.

14 **[Intermediate]** • **Target: Extrapolation**

Level 2 · MediumLet X and Y be independent random variables with $\sigma_X = 6$ and $\sigma_Y = 8$. What is the standard deviation of the difference $D = X - Y$?

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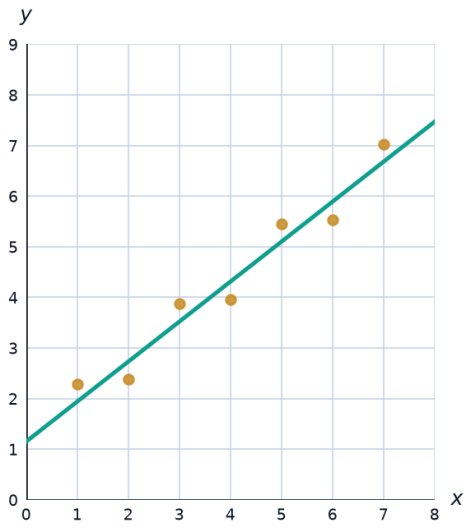
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging] • Target: Describing shape, center, and spread • Cross-Lesson: 3.03 The Normal Distribution**

[Unit 3 · Probability, Random Variables & Distributions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

16 **[Challenging] • Target: Scatterplots and correlation • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem**

Practice 4.2: Bivariate data for study hours x and exam mark y across 10 students yield $\bar{x} = 14.5$, $\bar{y} = 72.0$, and Pearson's correlation coefficient $r = 0.88$. The regression line of y on x has equation $y = 2.4x + c$. (a) Find the value of c. (b) Interpret the slope 2.4 in context. (c) Estimate the exam mark of a student who studies 18 hours.



17 **[Challenging] • Target: Interpreting the correlation coefficient • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing**

Level 2 · MediumTwo data sets have the same mean of 50 and the same sample size of 100. Dataset A has a standard deviation of 3.2, and Dataset B has a standard deviation of 8.9. Which statement must be true? (A) Dataset A values are more spread out from the mean than Dataset B values. (B) Dataset B has greater variability about the mean than Dataset A. (C) Both datasets have identical minimum and maximum values. (D) Dataset B has more outliers than Dataset A.

18 **[Challenging] • Target: Least-squares regression and residuals • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference**

A random sample of 200 students is classified by Grade Level (9th vs 12th) and Smartphone Operating System (iOS vs Android). Observed counts: 9th Grade: 70 iOS, 30 Android (100 total); 12th Grade: 50 iOS, 50 Android (100 total). (a) Calculate the expected count of 9th graders with iOS. (b) Compute the Chi-square test statistic χ^2. (c) With $df = 1$ and $\alpha = 0.05$ (critical value 3.84), conclude whether Grade Level and Operating System are independent.

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19 [Challenging] • Target: Coefficient of determination • Cross-Lesson: 1.01 Measures of Center & Spread

In the tomato experiment in Problem 7, the agronomist has two distinct tomato varieties: Roma and Beefsteak. (a) Explain why a randomized block design is superior to a completely randomized design. (b) Outline the blocking protocol.

20 [Challenging] • Target: Influential points and outliers in regression • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

A coffee brand claims its jars contain a mean of 200 grams of coffee. A consumer agency takes an SRS of 16 jars, finding $\bar{x} = 197.5$ g and $s = 4.0$ g. Assume jar weights are approximately normally distributed. Conduct a significance test of $H_0 : \mu = 200$ vs. $H_a : \mu < 200$ at $\alpha = 0.05$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Describing shape, center, and spread

A least-squares regression line is $\hat{y} = 16 - 5x$. Predict $\hat{y}$ when $x = 12$.

a. 76

b. -46

c. -42

d. -44

22 [Foundational] • Target: Scatterplots and correlation

A least-squares regression line is $\hat{y} = 52 + 2x$. For a point with $x = 2$ and observed value $y = 49$, find the residual (observed $-$ predicted).

a. -9

b. -5

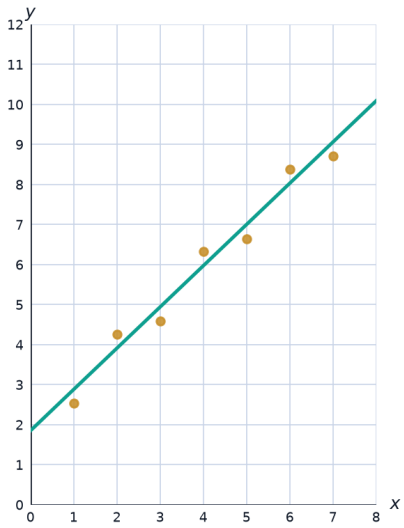
c. 7

d. -7

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23 [Foundational] • Target: Interpreting the correlation coefficient

A regression has correlation coefficient $r = -0.85$. Find r^2, expressed as a percentage, to two decimal places.



a. 74.25%

b. 85.00%

c. 70.25%

d. 72.25%

24 [Foundational] • Target: Least-squares regression and residuals

A distribution has mean 34 and median 47. Based on this comparison, classify the shape of the distribution.

a. Skewed right (mean noticeably greater than median)

b. Cannot be determined from the mean and median alone

c. Skewed left (mean noticeably less than median)

d. Approximately symmetric (mean and median are close)

25 [Foundational] • Target: Coefficient of determination

A regression has coefficient of determination $r^2 = 0.49$. Find the possible value(s) of the correlation coefficient r.

a. ± 0.80

b. ± 0.70

c. 0.70

d. ± 0.49

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26 [Foundational] • Target: Influential points and outliers in regression

A social scientist is investigating the relationship between the number of hours spent studying per week (X) and the final exam score (Y) for a group of $n = 100$ students. The least-squares regression line is given by $\hat{y} = 55.2 + 8.5x$. The standard error of the slope is $SE_b = 1.2$. The researcher wants to construct a 95% confidence interval for the true slope of the regression line. What is the value of the t-score used for this confidence interval?

a. 1.645

b. 1.960

c. 1.984

d. 2.576

27 [Foundational] • Target: Extrapolation

The correlation coefficient between two variables is $r = -0.31$. What percent of the variation in y is explained by the linear relationship with x?

a. 9.6\%

b. 0.5\%

c. 19.6\%

d. 29.6\%

28 [Foundational] • Target: Describing shape, center, and spread

The least-squares regression line for a data set is $\hat{y} = 11.1 + 2.5x$. What is the predicted value of y when $x = 45$?

a. 128.6

b. 123.6

c. 118.6

d. 125.6

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Scatterplots and correlation

A scatterplot is shown below. What is the approximate value of the correlation coefficient, r, for the data presented in the scatterplot?

a. -0.9

b. -0.2

c. 0.2

d. 0.9

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30 [Intermediate] • Target: Interpreting the correlation coefficient

In a regression analysis with $n = 23$ data points, the slope estimate is $b_1 = 2.31$ with standard error $SE_{b_1} = 0.909$. Find the test statistic t for testing $H_0 : \beta_1 = 0$.

a. 1.04

b. 4.54

c. 1.54

d. 2.54

31 [Intermediate] • Target: Least-squares regression and residuals

The coefficient of determination R^2 measures:

a. The slope

b. Variation in y explained by x

c. Correlation between residuals

d. Standard error of slope

32 [Intermediate] • Target: Coefficient of determination

Correlation coefficient r=–0.9 indicates:

a. Weak negative association

b. Strong positive association

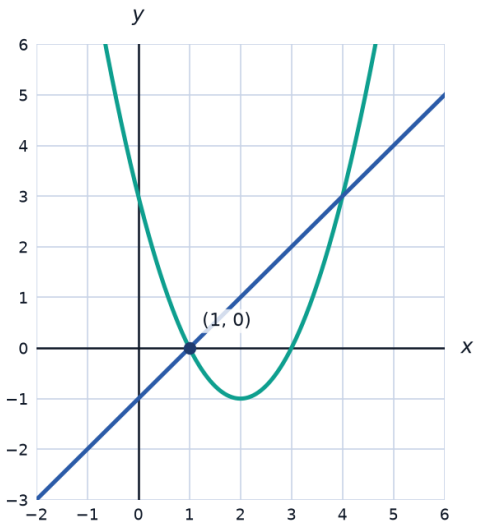
c. Strong negative association

d. No association

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33 **[Intermediate]** • **Target: Influential points and outliers in regression**

Consider the following diagram representing a probability space where events A and B are shown. If $P(A) = 0.6$ and $P(A \cap B) = 0.2$, and the area of the entire sample space is 1, what is the value of $P(B \setminus A)$?



a. 0.2

b. 0.4

c. 0.8

d. 0.6

34 **[Intermediate]** • **Target: Extrapolation**

A national polling organization is conducting a survey to estimate the proportion of registered voters in a large metropolitan area who support a particular ballot initiative. They plan to use a 95% confidence interval for the population proportion. The organization wants to ensure that the margin of error for this estimate does not exceed 2.5 percentage points. If they anticipate that the sample proportion will be close to 0.5 (the most conservative estimate for planning purposes), what is the minimum sample size, n, they must obtain to achieve this margin of error?

a. 1536

b. 1537

c. 1535

d. 1534

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35 [Intermediate] • Target: Describing shape, center, and spread

A discrete random variable, X, represents the number of successful trials in a sequence of independent Bernoulli trials. The probability of success on any given trial is $p = 0.6$. The expected value of X is $E[X] = 18$. We are interested in a new random variable, Y, defined as $Y = aX + b$, where a and b are constants. If the variance of Y is $Var(Y) = 7.2$, what is the value of a^2?

a. 0.16

b. 0.36

c. 1.00

d. 2.50

36 [Intermediate] • Target: Scatterplots and correlation

A bag contains marbles of different colors. The probability of drawing a red marble is $P(\text{Red}) = \frac{2}{5}$ and the probability of drawing a blue marble is $P(\text{Blue}) = \frac{1}{3}$. If the draws are independent, what is the probability of drawing a red marble followed by a blue marble?

a. $\frac{2}{15}$

b. $\frac{3}{8}$

c. $\frac{1}{2}$

d. $\frac{11}{15}$

37 [Intermediate] • Target: Interpreting the correlation coefficient

A discrete random variable X can take on the values $1, 2, 3,$ and 4. The probability mass function $P(X = x)$ is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. What is the value of k?

a. 1/10

b. 1/4

c. 1/6

d. 1/16

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Least-squares regression and residuals • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A data set consists of the following five points: $P_1 = (2, 3), P_2 = (4, y), P_3 = (6, 7), P_4 = (8, 9), P_5 = (10, 11)$. If the mean of the x-coordinates is 6 and the mean of the y-coordinates is 7, what is the value of y?

a. 3

b. 4

c. 5

d. 6

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39 [Challenging] • Target: Coefficient of determination • Cross-Lesson: 1.01 Measures of Center & Spread

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

a. 1/6

b. 1/3

c. 1/2

d. 2/3

40 [Challenging] • Target: Influential points and outliers in regression • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

Expected value E(X) for discrete X:

a. The most frequent value

b. $\sum x_i P(x_i)$

c. The median

d. $n \cdot p$ always

41 [Challenging] • Target: Extrapolation • Cross-Lesson: 2.01 Sampling Methods & Bias

If P(A)=0.4 and P(B|A)=0.5, then P(A∩B)=

a. 0.2

b. 0.5

c. 0.4

d. 0.9

42 [Challenging] • Target: Describing shape, center, and spread • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

A binomial random variable has $n = 25$ and $p = 0.64$. Find its mean $\mu = np$.

a. 13

b. 14

c. 16

d. 18

43 [Challenging] • Target: Scatterplots and correlation • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

If $P(A) = 0.56$, find $P(A^c)$ (the complement of A).

a. 0.44

b. 0.54

c. 0.64

d. 0.24

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44 [Challenging] • Target: Interpreting the correlation coefficient • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A geometric random variable has success probability $p = 0.48$. Find the mean (expected number of trials until first success), $\mu = 1/p$.

a. 2.08

b. 8.08

c. 4.08

d. 5.08

45 [Challenging] • Target: Least-squares regression and residuals • Cross-Lesson: 3.03 The Normal Distribution

In a bivariate regression with sample size $n = 31$ (Research Model 1), sample slope is $b_1 = 1.70$ with $\text{SE}_{b_1} = 0.26$. Testing $H_0 : \beta_1 = 0$, what is the t-test statistic and degrees of freedom?

a. $t = 6.54$, df $= 29$

b. $t = 0.44$, df $= 29$

c. $t = 8.08$, df $= 31$

d. $t = 6.54$, df $= 30$

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